

# From Surcomplex Hahn Fields to Berkovich Geometry

Rank-one convergence, Tate algebras,  
analytic annuli, and an entire-function case study

Research exposition prepared for Vladimir Reshetnikov  
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## Abstract

The repository develops several support-controlled analytic categories over surreal and surcomplex numbers, and briefly compares them with Berkovich geometry. This article fills a different, more concrete gap: it constructs a rank-one Banach-analytic theory inside the surcomplex field and proves exactly how it relates to the repository's Hahn-coefficient rings.

Working primarily in  $K = \mathbb{C}((t^{\mathbb{R}})) \subset \mathbf{No}[i]$ , we distinguish intrinsic valuation convergence from strong Hahn summability and the full surreal fine topology. We prove strict comparisons of the function rings, construct one-variable Weierstrass preparation by a contraction argument, and derive analytic root counts and a valuation Rouché theorem. We then build the relevant Berkovich points, explain disks, branches and annulus skeletons, and calculate analytic de Rham cohomology and residues without pretending that an ultrametric annulus has an ordinary circular contour.

The case study  $F(Z) = \sum_{n \geq 0} t^{n^2} Z^n$  has exactly one simple zero of valuation  $-(2m-1)$  for every  $m \geq 1$ , all its zeros are real and negative, and their first corrections and a convergent canonical product are explicit. This example also locates the precise boundary between fixed-workspace entireness and an assertion on the whole proper class  $\mathbf{No}[i]$ .

**Status and scope.** A theorem-driven exposition and comparison, not a claim of first occurrence or the solution of a named open problem. Deep imported results are identified; the central comparison and computational examples are proved here. The general arguments are not formally verified. The accompanying program checks 542 finite exact identities and coefficient assertions.

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# 1 The documentation gap and the mathematical contract

## 1.1 What the repository already contains

The original audit behind this article used the repository snapshot

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from the default branch, inspected on September 21, 2026. It was a targeted review of the catalogue, the relevant report READMEs, and the introductory and comparison material in the analytic reports, not a line-by-line certification of every manuscript or a lexical search asserted to be exhaustive.

The distinction matters because this is not an empty subject in the repository. The *analysis* report develops infinitesimal and Hahn-coherent function theory. The *analytic-geometry* report distinguishes ordinary convergent germs, common-domain coefficient families, and formal coefficients. The *polynomial-algebra* report gives extensive finite-degree valuation geometry. Finally, *contours-and-stokes* explicitly compares its integration categories with Berkovich and tropical approaches [5, 6, 7, 8, 9].

At that snapshot the identified gap was *an undeveloped bridge*, rather than the absence of the word “Berkovich”. The inspected reports did not provide a sustained construction of a rank-one Tate algebra, identify its exact position among their coefficient rings, and then work through the associated Berkovich disks, analytic annuli and a nonpolynomial entire function in that category. Those are the tasks undertaken here. The existing discussions of the intrinsic topology of a Hahn workspace already avoid identifying it with the topology induced from all of  $\mathbf{No}[i]$ ; we amplify that correct distinction rather than attribute an error to them.

| Existing report     | Existing emphasis                                                              | Added here                                                               |
|---------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Analysis            | Strong summability and coherent ordinary coefficient functions                 | A complete real-valued norm and actual Banach convergence                |
| Analytic geometry   | Radius-free and common-domain Hahn germ rings                                  | An explicit strict comparison with Tate and open-disk algebras           |
| Polynomial algebra  | Finite-degree Newton profiles and valuation balls                              | Preparation, root counts and slopes for convergent infinite series       |
| Contours and Stokes | Coefficientwise, deformed and algebraic contour pairings; category comparisons | Berkovich disk points, annulus skeletons and analytic Laurent primitives |
| Foundations         | Fixed-workspace versus proper-class scope; a quadratic-exponent entire example | The complete zero spectrum and product expansion of that example         |

## 1.2 Three questions that must be answered separately

For a displayed infinite expression, the following are different questions. Does it define a Hahn sum? Do its partial sums converge in the intrinsic valuation topology of a specified field? Do they converge when every surreal scale is allowed as a test of smallness? A positive answer to the first or the second does not give a positive answer to the third.

Likewise a function ring, a topology on its points, and an integration theory are separate choices. Our primary objects are Banach algebras over a fixed complete rank-one valued field. Their spectra are set-sized spaces of seminorms. The associated differential is differentiation in a coordinate  $Z$ , fixing the coefficient field. None of these conventions selects a derivation of the surreal numbers themselves or defines a contour integral on arbitrary fine-topological surcomplex paths.

### 1.3 Established inputs and the level of proof

We use the standard Hahn-field construction and the algebraic closedness theorem for divisible value group and algebraically closed residue field; Poonen’s Corollary 4 supplies a precise reference [1]. The general point classification and the annulus deformation retraction in Berkovich geometry are imported from the standard theory [2, 3]. In contrast, the support and convergence comparisons, the one-variable preparation argument, root-count formulas, residue computations, and the entire-function case study are given with proofs in this article.

These are adaptations and deductions in an established subject, not a priority claim based on replacing “non-Archimedean” by “surcomplex”. In particular, the worked series is already used in the repository’s foundations discussion, and is a rescaling of the classical partial theta series [10, 4]. What is supplied here is a unified valuation-analytic treatment with its hypotheses exposed and its finite calculations reproducible.

## 2 A complete rank-one field inside the surcomplex numbers

### 2.1 Normal forms and the two meanings of size

For a set-sized ordered additive subgroup  $\Gamma \subset \mathbf{No}$ , write

$$K_\Gamma = \mathbb{C}((t^\Gamma)), \quad t^\gamma := \omega^{-\gamma}.$$

The last expression is Conway’s monomial map. It is not an instruction to use Gonshor exponentiation with a general surreal exponent. An element is a formal sum

$$a = \sum_{\gamma \in S} c_\gamma t^\gamma, \quad c_\gamma \in \mathbb{C}^*, \quad S \subset \Gamma \text{ well ordered.} \quad (2.1)$$

The zero element has empty support. All supports and all function algebras used below are sets. Hahn addition and multiplication identify this field with a subfield of  $\mathbf{No}[i]$ , with conjugation on the coefficients [6, 1].

From now on the main field is

$$K = \mathbb{C}((t^{\mathbb{R}})), \quad v(a) = \min \text{supp}(a), \quad |a|_v = e^{-v(a)} \quad (a \neq 0). \quad (2.2)$$

We put  $v(0) = +\infty$  and  $|0|_v = 0$ . Then

$$|ab|_v = |a|_v |b|_v, \quad |a + b|_v \leq \max\{|a|_v, |b|_v\}.$$

The real exponential in (2.2) merely turns a real-valued valuation into a positive real-valued norm. It is *not* a specialization of the surreal number  $t$  to the real number  $e^{-1}$ .

In particular,

$$|c|_v = 1 \quad (c \in \mathbb{C}^*), \quad |n|_v = |n!|_v = 1 \quad (n \in \mathbb{N}_{>0}). \quad (2.3)$$

The ordinary complex absolute value of a coefficient is irrelevant to this norm. We reserve  $|\cdot|_v$  for the valuation norm throughout. The surreal-valued modulus  $\sqrt{x^2 + y^2}$  is a different object. Its unit disk, for example, is not the valuation unit disk: every nonzero ordinary complex number has valuation norm one.

The valuation ring, maximal ideal and residue field are

$$\mathcal{O}_K = \{a : v(a) \geq 0\}, \quad \mathfrak{m}_K = \{a : v(a) > 0\}, \quad \mathcal{O}_K/\mathfrak{m}_K \simeq \mathbb{C}.$$

Reduction  $\text{red}$  extracts the coefficient of  $t^0$  on  $\mathcal{O}_K$ . Every positive real radius occurs: if  $R = e^{-\beta}$ , then  $|t^\beta|_v = R$ . The algebraic closedness of  $K$  follows from the cited Hahn-field theorem because  $\mathbb{R}$  is divisible and  $\mathbb{C}$  is algebraically closed. Its real-coefficient subfield  $\mathbb{R}((t^{\mathbb{R}}))$  is consequently real closed.

## 2.2 Why this Hahn field is spherically complete

**Definition 2.1.** A valued field is *spherically complete* if every nonempty nested family of nonempty closed valuation balls has nonempty intersection.

**Proposition 2.2.** *The field  $K = \mathbb{C}((t^{\mathbb{R}}))$  is spherically complete, and in particular complete for the metric  $d(a, b) = |a - b|_v$ .*

*Proof.* A ball of positive radius has the form

$$B_i = \{x : v(x - a_i) \geq \gamma_i\}.$$

It prescribes exactly the coefficients of  $x$  at exponents strictly below  $\gamma_i$ . In a nested family these prescriptions agree wherever both apply: if one ball is contained in another, its center belongs to the larger ball. Glue all prescribed coefficients, and set every coefficient not prescribed to zero.

The support of the glued expression is well ordered. Indeed, let  $T$  be a nonempty subset of that support and choose  $s \in T$ . Some prescription contains the coefficient at  $s$ , so  $s < \gamma_i$  for one  $i$ . All the prescribed coefficients at exponents at most  $s$  agree with  $a_i$ : a smaller ball agrees below  $\gamma_i$ , and a larger ball's prescription agrees with that of  $a_i$  where it is defined. Therefore  $T \cap (-\infty, s]$  is a nonempty subset of the well-ordered support of  $a_i$ . Its least element is also the least element of  $T$ .

The glued expression is a Hahn series and satisfies every prescription, hence belongs to every ball. A ball of radius zero is a singleton and causes no problem. Finally a Cauchy sequence has a subsequence contained successively in nested balls with radii tending to zero; an intersection point is its limit and then the limit of the original sequence.  $\square$

This proof uses the full Hahn field, not the completion of finite monomial expressions. Those need not coincide. For example,

$$h = \sum_{n \geq 0} t^{1-1/(n+1)}$$

is a Hahn series, but no finite-support series approximates it with error valuation greater than one. The infinitely many nonzero coefficients below one would all have to be matched. In  $\mathbb{C}((t^{\mathbb{Q}}))$ , the metric closure of the finite-support series consists precisely of the series whose support has finite intersection with  $(-\infty, B]$  for every real  $B$ : necessity follows from approximation to each cutoff, and sufficiency from finite truncation. The full Hahn field is larger. Spherical completeness should never be inferred merely from the word “completion”.

## 2.3 Intrinsic topology versus full surreal topology

**Proposition 2.3.** *In the intrinsic valuation topology of  $K$ ,  $t^n \rightarrow 0$ . The subspace topology that  $K$  inherits from the full fine topology of  $\mathbf{No}[i]$  is discrete, and in that topology  $t^n$  does not converge to zero.*

*Proof.* Intrinsically  $|t^n|_v = e^{-n} \rightarrow 0$ . In the full surreal field, however, the positive radius  $t^\omega$  is smaller than the surreal modulus of every nonzero element of  $K$ : the latter has a real leading exponent, whereas  $\omega$  exceeds every real exponent. Thus the fine ball of radius  $t^\omega$  around an element of  $K$  meets  $K$  only at that element. In particular none of the  $t^n$  belongs to the fine ball of that radius about zero.  $\square$

The distinction is a change in topology, not a conflict about the field operations. The detailed contour report already states the corresponding workspace qualification [9]. A theorem about fine convergence on the full proper class is not a theorem about this intrinsic metric.

### 3 Convergence, summability, and the rank-one boundary

#### 3.1 When an infinite sum really is a limit

A family of Hahn series is *strongly summable* if the union of its supports is well ordered and only finitely many members contribute to any specified exponent. Its sum is then defined coefficientwise. This definition has no reference to a norm limit.

**Theorem 3.1.** *For a sequence  $(a_n)$  in  $K$ , the following are equivalent:*

$$\sum_{n \geq 0} a_n \text{ converges in the intrinsic norm,} \quad |a_n|_v \longrightarrow 0, \quad v(a_n) \longrightarrow +\infty.$$

*When they hold, the family  $(a_n)$  is strongly summable and its Hahn sum is the norm limit. Strong summability alone does not imply any of these conditions.*

*Proof.* Convergence forces the differences of consecutive partial sums to tend to zero. Conversely, the ultrametric inequality bounds every finite tail by the largest norm of a term in that tail. Terms tending to zero therefore make the partial sums Cauchy; Proposition 2.2 supplies their limit.

Fix an exponent cutoff  $B$ . Only finitely many  $a_n$  have valuation at most  $B$ , so the part of the support union at or below  $B$  is contained in a finite union of well-ordered sets. To find the least element of an arbitrary nonempty subset of the union, choose any one of its elements and use it as this cutoff. This proves well ordering. A fixed exponent occurs in only finitely many members for the same reason. Each coefficient stabilizes in the partial sums, and the stabilized coefficients give the norm limit.

For the failure of the converse take  $a_n = t^{1-1/(n+1)}$ . Its supports form a well-ordered increasing sequence, with no repeated exponent, but  $|a_n|_v \rightarrow e^{-1}$  rather than zero.  $\square$

There is a second useful warning in this example: a single Hahn coefficient may have infinitely many terms below a finite cutoff. Banach convergence of a sequence of such coefficients does not turn every coefficient into a finite object or provide an algorithm for accessing its support.

#### 3.2 Why higher rank cannot be silently scalarized

**Proposition 3.2.** *The ordered group  $\mathbb{Q} + \mathbb{Q}\omega \subset \mathbf{No}$  has no order-preserving injective additive map to  $\mathbb{R}$ . In particular, its natural valuation cannot be converted faithfully to a real-valued absolute value by an order embedding.*

*Proof.* If  $\phi$  were such an embedding,  $\phi(1) > 0$ . But  $n < \omega$  for every positive integer  $n$ , so  $n\phi(1) < \phi(\omega)$  for every  $n$ , contrary to the Archimedean property of  $\mathbb{R}$ .  $\square$

A noninjective order-preserving homomorphism can discard an entire layer of scale; in this example it must kill 1. Such a coarsening can be useful, but is not a faithful replacement of the original valuation. The sequence  $t^n$  again diagnoses the issue: its valuations are unbounded in  $\mathbb{R}$ , but are all below  $\omega$  in  $\mathbb{Q} + \mathbb{Q}\omega$ .

The repository's localization principle puts any given *set* of surcomplex numbers in some set-sized divisible Hahn workspace [10]. It does not put every such set in a workspace whose natural value group has rank one. Already the pair  $t, t^\omega$  prevents that conclusion. Our results therefore concern a substantial subfield, not an analytic structure transported without loss to every surcomplex scale.

The arbitrary-rank case is taken up elsewhere in the collection, on a different object, by the report on Hahn–Tate uniformization [11]. It keeps the full ordered group  $\Gamma$ , restricts the

multiplicative Tate parameter to the principal convex subgroup generated by  $v(q)$ , and applies a real-valued coarsening only inside that subgroup, where  $v(q)$  is an order unit; the scales killed by the coarsening are recovered there by formal evaluation rather than by a norm. A second proof in that report (`tate:node:thm:main`) uses no real-valued coarsening at all: it inverts the parametrization by integral formal charts over any coefficient field of characteristic zero. It cites Proposition 3.2 above as the reason that higher rank cannot instead be scalarized. A Tate algebra is not a Tate elliptic curve: that report does not reprove, extend, or supersede anything established here, and nothing here is asserted about elliptic curves.

### 3.3 Legitimate changes of workspace

The inclusions

$$\mathbb{C}((t^{\mathbb{Q}})) \subset \mathbb{C}((t^{\mathbb{R}})) \subset \mathbf{No}[i]$$

have different analytic meanings. The first is isometric for the displayed real-valued norms. A norm-convergent sequence over the smaller field retains its limit in the larger rank-one field. The second has no such implication when the topology on the target is the full fine topology. Conversely, enlarging  $\mathbb{Q}$  to  $\mathbb{R}$  fills out the set of available valuation radii; it changes the types of nonclassical disk points discussed in Section 7.

## 4 Tate algebras and closed valuation disks

### 4.1 The coefficient condition is the analytic definition

Fix  $R > 0$  and write  $R = e^{-\beta}$ ,  $\beta \in \mathbb{R}$ . The closed valuation disk and its Tate algebra are

$$D_R(K) = \{z \in K : |z|_v \leq R\}, \quad (4.1)$$

$$A_R = K\langle R^{-1}Z \rangle = \left\{ \sum_{n \geq 0} a_n Z^n : |a_n|_v R^n \longrightarrow 0 \right\}. \quad (4.2)$$

The notation  $R^{-1}Z$  records the norm assigned to the coordinate; it does not regard  $R$  as a coefficient with that valuation norm. Substitution  $Z = t^{\beta}U$  identifies  $A_R$  isometrically with the unit-radius algebra  $\mathcal{T} = K\langle U \rangle$ . Equivalently, the condition in (4.2) is

$$v(a_n) + n\beta \longrightarrow +\infty. \quad (4.3)$$

The Gauss norm is

$$\|f\|_R = \max_{n \geq 0} |a_n|_v R^n.$$

The maximum exists for  $f \neq 0$  because the terms tend to zero. An infinite formal expression without the tail condition is not a member of  $A_R$  merely because each of its coefficients lies in  $K$ .

**Proposition 4.1.** *The algebra  $A_R$  is a complete normed  $K$ -algebra; polynomials are dense; evaluation at  $z \in D_R(K)$  is defined by a norm-convergent series and satisfies  $|f(z)|_v \leq \|f\|_R$ . The Gauss norm is multiplicative.*

*Proof.* The weighted coefficients of a Cauchy sequence converge coordinatewise in  $K$ . To check the vanishing-tail condition for its limit, approximate the sequence uniformly by one member and then discard that member's sufficiently far tail. This also proves convergence in Gauss norm. Polynomial truncation proves density. Evaluation and its bound follow from Theorem 3.1 and the ultrametric inequality. Multiplication is the usual Cauchy product, justified either by the same estimates or by continuous extension from polynomials.

For multiplicativity reduce to  $R = 1$ . Divide each nonzero factor by one of its coefficients of maximal norm. The normalized series have all coefficients in  $\mathcal{O}_K$ , and their reductions are nonzero polynomials in  $\mathbb{C}[U]$ : only finitely many coefficients have norm one. Reduction of the product is the product of these two polynomials, which is nonzero. The product consequently has norm one, as claimed after reversing the normalizations.  $\square$

**Proposition 4.2.** *For every  $f \in A_R$ ,*

$$\|f\|_R = \max_{z \in D_R(K)} |f(z)|_v.$$

*Proof.* For  $f = 0$  the assertion is immediate. Otherwise normalize and scale as in the preceding proof. Choose  $c \in \mathbb{C}$  which is not a root of the nonzero reduction polynomial. Evaluation at  $U = c$  then has norm one. Such a  $c$  exists because  $\mathbb{C}$  is infinite. Rescaling proves the result.  $\square$

This is a maximum-modulus statement for a valuation norm. It neither selects an ordinary circular contour nor implies that all functions attain their maxima at one classical point. A single *nonclassical* point will do that in the Berkovich spectrum.

## 4.2 Two familiar series with different behavior

The geometric series  $\sum_{n \geq 0} t^n Z^n$  belongs to  $A_1$  and is the inverse of  $1 - tZ$  there. Its equality with that inverse is both an algebraic identity and an intrinsic norm limit. In contrast,

$$\sum_{n \geq 0} \frac{Z^n}{n!}$$

does not belong to  $A_1$ , since every coefficient has norm one. It does belong to  $A_R$  for every  $R < 1$ . Ordinary complex entireness of the coefficient function  $e^Z$  and valuation-analytic entireness in the variable  $Z$  are thus entirely different requirements.

# 5 An exact bridge between analytic function rings

## 5.1 The rings and the common ambient formal algebra

Use the same variable  $Z$  throughout and put

$$\begin{aligned} \mathcal{T} &= K\langle Z \rangle, & \mathcal{P} &= \mathbb{C}[Z]((t^{\mathbb{R}})), \\ \mathcal{A} &= \mathbb{C}\{Z\}((t^{\mathbb{R}})), & \mathcal{F} &= \mathbb{C}[[Z]]((t^{\mathbb{R}})). \end{aligned}$$

Here  $\mathbb{C}\{Z\}$  is the ordinary complex ring of convergent germs. In  $\mathcal{A}$  each Hahn coefficient is such a germ, with no common complex radius required. The symbols  $\mathcal{P}, \mathcal{A}, \mathcal{F}$  demand one well-ordered Hahn support for the whole family; they are not arbitrary collections of coefficients in  $K$ .

All four rings map into  $K[[Z]]$ . To describe the map explicitly write

$$H = \sum_{s \in S} h_s(Z) t^s, \quad h_s(Z) = \sum_{n \geq 0} c_{s,n} Z^n, \quad b_n = \sum_{s \in S} c_{s,n} t^s.$$

Then  $H$  maps to  $\sum b_n Z^n$ . Each  $b_n$  has support contained in  $S$ ; coefficient uniqueness makes the map injective. The usual finite decomposition property for sums of well-ordered supports makes it a ring homomorphism. These are coefficientwise identities, not a Fubini theorem for arbitrary norm limits.

Finally define the analytic open-unit-disk ring by

$$\mathcal{E}^- = \left\{ \sum_{n \geq 0} b_n Z^n \in K[[Z]] : v(b_n) + n\varepsilon \rightarrow +\infty \text{ for every } \varepsilon > 0 \right\}. \quad (5.1)$$

It is equivalently the ring of compatible power-series sections on all centered closed disks of radius less than one. Such disks exhaust the open disk. This explicit description suffices here; the usual Berkovich analytic interpretation agrees with it [3, Section I.4].

## 5.2 The strict comparison theorem

**Theorem 5.1.** *Under the preceding coefficient embeddings, the following inclusions of  $K$ -algebras inside  $K[[Z]]$  are all strict:*

$$K[Z] \subsetneq \mathcal{T} \subsetneq \mathcal{P} \subsetneq \mathcal{A} \subsetneq \mathcal{F} \subsetneq \mathcal{E}^-. \quad (5.2)$$

Moreover, membership in  $\mathcal{F}$  is equivalent to the union  $\bigcup_n \text{supp}(b_n)$  being well ordered. Within  $\mathcal{F}$ , membership in  $\mathcal{P}$  means that every exponent-row  $(c_{s,n})_{n \geq 0}$  has finite support, and membership in  $\mathcal{A}$  means that every such row is an ordinary convergent germ.

*Proof.* The criterion for  $\mathcal{F}$  follows directly by transposing the coefficient matrix; its support is exactly the union of the nonzero column supports. The additional row conditions are precisely the definitions of  $\mathcal{P}$  and  $\mathcal{A}$ .

For  $\mathcal{T} \subset \mathcal{P}$ , use  $v(b_n) \rightarrow +\infty$ . At any fixed exponent only finitely many columns can contribute, so each row is a polynomial. Below any fixed exponent cutoff, only finitely many column supports occur. Their union is well ordered; the least-element argument in Theorem 3.1 shows that the full union is well ordered as well.

For  $\mathcal{F} \subset \mathcal{E}^-$ , a nonzero common support  $S$  has a least exponent  $s_0$ . Thus  $v(b_n) \geq s_0$  for every  $n$ , and  $v(b_n) + n\varepsilon \geq s_0 + n\varepsilon \rightarrow +\infty$ . The remaining inclusions are immediate from the coefficient definitions.

The successive strictness witnesses are as follows:

$$\begin{aligned} \mathcal{T} \setminus K[Z] &: \sum_{n \geq 0} t^n Z^n, \\ \mathcal{P} \setminus \mathcal{T} &: \sum_{n \geq 0} t^{1-1/(n+1)} Z^n, \\ \mathcal{A} \setminus \mathcal{P} &: \sum_{n \geq 0} Z^n / n!, \\ \mathcal{F} \setminus \mathcal{A} &: \sum_{n \geq 0} n! Z^n, \\ \mathcal{E}^- \setminus \mathcal{F} &: \sum_{n \geq 0} t^{-\sqrt{n}} Z^n. \end{aligned}$$

The second has polynomial exponent-rows but column valuations tending to one, not to infinity. The third and fourth have the single Hahn exponent zero: their coefficient functions are respectively a nonpolynomial convergent germ and a divergent formal series. In the last,  $n\varepsilon - \sqrt{n} \rightarrow +\infty$  for every positive  $\varepsilon$ , but the union of coefficient supports is unbounded below and hence not well ordered.  $\square$

This theorem is the central bridge missing from a mere catalogue of analytic categories. In particular,  $\mathbb{C}[Z]((t^{\mathbb{R}}))$  is *not* the unit-disk Tate algebra, even though both may be described informally as series with polynomial coefficient data. A common Hahn support is a different condition from a Gauss-norm tail tending to zero.

### 5.3 Where the common-domain rings fit

The two common-domain categories of the repository fit into this comparison as well. Let  $D$  be an ordinary complex open disk centered at zero, and put

$$\mathcal{H}(D) = \mathcal{O}(D)((t^{\mathbb{R}})), \quad \mathcal{R}_1 = \varinjlim_{D \ni 0} \mathcal{H}(D).$$

The second ring permits one ordinary domain for the entire coefficient family, followed by shrinking it; it does not permit a different domain for each coefficient. Taking coefficient Taylor germs gives injections into  $\mathcal{A}$ . For  $\mathcal{H}(D)$  injectivity uses the ordinary identity theorem on the connected disk, coefficient by coefficient.

There is the further strict refinement

$$\mathcal{P} \subsetneq \mathcal{H}(D) \subsetneq \mathcal{R}_1 \subsetneq \mathcal{A}. \quad (5.3)$$

Every polynomial coefficient is holomorphic on  $D$ , while the single coefficient  $e^Z$  witnesses the first strictness. For the second, choose  $0 \neq b \in D$ ; the germ of  $1/(Z - b)$  is in  $\mathcal{R}_1$  but cannot extend holomorphically across  $b$  on  $D$ . For the third, use  $\sum_{j \geq 1} t^j/(1 - jZ)$ : its individually convergent coefficient germs have poles at  $1/j$ , so no common ordinary neighborhood works. Uniqueness of Hahn coefficients prevents a different representative from removing these poles. Thus (5.2) and (5.3) locate both common-domain rings without identifying either of them with the Tate algebra.

### 5.4 Positive rescaling makes the bridge operational

**Corollary 5.2.** *For every  $\alpha > 0$ , substitution  $Z = t^\alpha U$  gives an injective  $K$ -algebra homomorphism*

$$\mathcal{F} \longrightarrow K\langle U \rangle.$$

*Consequently every element of  $\mathcal{A}$ , including a family with no common ordinary complex radius, becomes a closed-unit-disk Tate function after positive monomial rescaling.*

*Proof.* With  $s_0$  as in the previous proof, the coefficient of  $U^n$  has valuation  $v(b_n) + n\alpha \geq s_0 + n\alpha \rightarrow +\infty$ . Formal substitution by a nonzero scalar is injective and respects products.  $\square$

The repository's contour theory uses positive rescaling to obtain polynomial-coefficient Hahn families [9]. In rank one the same lower bound gives the stronger, explicit Banach-tail conclusion above. For an arbitrary ordered group the step  $n\alpha \rightarrow +\infty$  need not hold; Corollary 5.2 is not being asserted unchanged in higher rank.

For example, the familiar radius-free witness

$$H(Z) = \sum_{j \geq 1} \frac{t^j}{1 - jZ}$$

has coefficient functions with ordinary poles approaching zero. Nevertheless  $H(t^\alpha U)$  belongs to  $K\langle U \rangle$  for every  $\alpha > 0$ . This does not remove those ordinary complex poles. It changes to an infinitesimal valuation disk and changes which notion of convergence is being asked about.

## 6 Preparation and zero counting for convergent infinite series

### 6.1 Division as a contraction rather than a formal slogan

The preparation theorem is the point at which an infinite analytic function acquires finite algebraic control on a fixed closed disk. We give a proof in one variable, keeping the norm

estimates visible. It is the usual non-Archimedean Weierstrass mechanism in a form suited to the chosen Hahn field, not an appeal to an unproved preparation theorem for one of the larger rings in Section 5.

**Lemma 6.1.** *Let  $p(U) \in \mathcal{O}_K[U]$  be monic of degree  $d \geq 0$ . Polynomial division by  $p$  extends to bounded  $K$ -linear maps*

$$Q_p : \mathcal{T} \longrightarrow \mathcal{T}, \quad R_p : \mathcal{T} \longrightarrow K[U]_{<d},$$

*with operator norms at most one and  $g = pQ_p(g) + R_p(g)$ . Here  $K[U]_{<0} = \{0\}$  when  $d = 0$ .*

*Proof.* For a polynomial, ordinary descending-degree division by a monic polynomial whose other coefficients have norm at most one never increases the largest coefficient norm. Thus its quotient and remainder are bounded by the norm of the dividend. Both operators extend continuously from the dense polynomials. The target of  $R_p$  is finite-dimensional and complete; the division identity passes to the limit. The same continuity argument gives  $Q_p(pq + r) = q$  and  $R_p(pq + r) = r$  for  $q \in \mathcal{T}$  and  $r \in K[U]_{<d}$ , first on polynomials and then by density. For  $d = 0$ ,  $p = 1$ , the quotient is  $g$  and the remainder zero.  $\square$

**Theorem 6.2** (One-variable Weierstrass division and preparation). *Let  $0 \neq f(U) = \sum a_n U^n \in \mathcal{T}$ , and let*

$$d = \max\{n : |a_n|_v = \|f\|_1\}.$$

*After dividing by  $a_d$ , suppose  $a_d = 1$  and  $\|f\|_1 = 1$ . For every  $g \in \mathcal{T}$  there are unique  $q \in \mathcal{T}$  and  $r \in K[U]_{<d}$  such that  $g = qf + r$ . Moreover*

$$f = uW, \tag{6.1}$$

*where  $W \in \mathcal{O}_K[U]$  is monic of degree  $d$  and  $u$  is a unit satisfying  $\|u - 1\|_1 < 1$ . Without the normalization, a nonzero scalar multiplies  $u$ .*

*Proof.* Write

$$p = \sum_{n=0}^d a_n U^n, \quad h = f - p, \quad \theta = \|h\|_1 < 1.$$

The strict inequality uses both the choice of the *last* maximal coefficient and the fact that the tail tends to zero. By Lemma 6.1, the equation

$$q = Q_p(g - qh) \tag{6.2}$$

is a contraction on the complete normed space  $\mathcal{T}$ , with contraction constant at most  $\theta$ . Its fixed point exists and is unique. Put  $r = R_p(g - qh)$ ; then  $g = q(p + h) + r = qf + r$ .

Conversely any such division must satisfy (6.2) upon applying  $Q_p$  to  $g - qh = pq + r$ . This proves uniqueness. Starting the iteration at zero also shows  $\|q\|_1, \|r\|_1 \leq \|g\|_1$ .

Apply this construction to  $g = U^d$ . Because  $Q_p(U^d) = 1$ , the fixed point obeys  $\|q - 1\|_1 \leq \theta$  and  $\|q\|_1 = 1$ . The geometric series shows that  $q$  is a unit with  $q^{-1} = 1 +$  a term of norm less than one. Now

$$W = U^d - r = qf$$

is monic of degree  $d$ , its coefficients have norm at most one, and  $f = q^{-1}W$  has the asserted form.  $\square$

Every invocation of a geometric series or a fixed-point limit in this proof is in the intrinsic Banach topology. In particular the proof cannot simply be relabelled as a fine-topological proof over all of  $\mathbf{No}[i]$ .

**Corollary 6.3.** *A series  $f \in \mathcal{T}$  is a unit if and only if*

$$|a_0|_v > \sup_{n \geq 1} |a_n|_v.$$

*Equivalently  $f = a_0(1 + h)$  with  $\|h\|_1 < 1$ .*

*Proof.* The displayed form is invertible by the convergent geometric series. In the other direction, normalize  $f$  and its inverse and reduce modulo  $\mathfrak{m}_K$ . Multiplicativity of the Gauss norm implies that their nonzero reduction polynomials multiply to a nonzero constant. Both reductions must be constants. The tail condition then makes the displayed inequality strict.  $\square$

## 6.2 Finite quotients and one-variable ideal theory

**Corollary 6.4.** *Every ideal of  $\mathcal{T} = K\langle U \rangle$  is principal. For nonzero  $f = uW$  as in preparation,*

$$\mathcal{T}/(f) \simeq K[U]/(W), \quad \dim_K \mathcal{T}/(f) = \deg W.$$

*Every maximal ideal is  $(U - a)$  for a unique  $a \in \mathcal{O}_K$ , and its residue field is  $K$ .*

*Proof.* In a nonzero ideal choose  $f$  with least possible Weierstrass degree. Divide any other member  $g$  by  $f$ . The remainder is again in the ideal and has polynomial degree less than that Weierstrass degree. If nonzero, its own Weierstrass degree is no larger than its polynomial degree, a contradiction. Thus every remainder is zero and  $f$  generates the ideal.

Division by the monic  $W$  gives unique polynomial representatives of degree less than  $\deg W$ . On polynomials analytic division agrees with ordinary polynomial division by uniqueness; this proves the asserted quotient isomorphism. A proper maximal ideal cannot have Weierstrass degree zero. Since  $K$  is algebraically closed, its prepared polynomial factors into linear terms with roots in  $\mathcal{O}_K$ , and maximality forces just one such factor. Distinct  $a$  give distinct evaluation ideals.  $\square$

Consequently a finite quotient of the disk algebra is available through ordinary multiplication matrices, including nonreduced quotients at multiple zeros. This is a precise one-variable analytic input to the repository's finite algebra methods. It proves no corresponding ideal theorem for an unnamed larger function ring merely because its elements also have Hahn coefficients.

There is also a useful warning about the word “local”. The function on  $\mathcal{O}_K$  which is zero on  $\mathfrak{m}_K$  and one on its complement is locally constant for the intrinsic ultrametric topology. It is thus locally represented by constant power series and has derivative zero at every classical point. It is not a member of  $\mathcal{T}$ : any nonzero Tate function has only finitely many zeros in the closed disk, by preparation and algebraic closedness, whereas this function has infinitely many zeros and is not identically zero. Arbitrary pointwise local power-series data in the naive topology are therefore a larger and less rigid category than functions in a disk Tate algebra.

## 6.3 Disk, shell and residue-direction counts

For  $0 \neq f(Z) = \sum a_n Z^n \in A_R$  set

$$\begin{aligned} \lambda_R &= \min_n (v(a_n) + n\beta), \\ m_R &= \min\{n : v(a_n) + n\beta = \lambda_R\}, \\ d_R &= \max\{n : v(a_n) + n\beta = \lambda_R\}, \end{aligned} \tag{6.3}$$

where  $R = e^{-\beta}$ . The normalized initial polynomial is

$$I_R(U) = \text{red} \left( t^{-\lambda_R} f(t^\beta U) \right) \in \mathbb{C}[U]. \tag{6.4}$$

It has degree  $d_R$  and order  $m_R$  at zero. The normalization is an actual monomial in  $K$ , not a norm viewed as a coefficient.

**Theorem 6.5.** *With multiplicities counted,  $f$  has exactly  $d_R$  zeros in  $D_R(K)$ , exactly  $m_R$  zeros in  $\{|z|_v < R\}$ , and exactly  $d_R - m_R$  zeros on the shell  $\{|z|_v = R\}$ . For each  $c \in \mathbb{C}^*$  the number in*

$$\left\{z \in D_R(K) : \text{red}(t^{-\beta}z) = c\right\}$$

*is the multiplicity of  $c$  as a root of  $I_R$ .*

*Proof.* Prepare the scaled series. Its unit factor has no zeros on the closed unit disk, so all zeros come from a degree- $d_R$  monic polynomial  $W$  whose coefficients have norm at most one. Every root of  $W$  has norm at most one: at a point of norm greater than one its leading monomial strictly dominates all other terms. Algebraic closedness supplies exactly  $d_R$  roots with multiplicity.

Factor  $W = \prod_j (U - u_j)$  in  $K[U]$ . The  $u_j$  lie in  $\mathcal{O}_K$ , so  $\text{red}(W) = \prod_j (U - \text{red}(u_j))$ . The reduction of the analytic unit is a nonzero constant, and consequently this polynomial differs from  $I_R$  by a nonzero constant. Roots  $u_j$  of norm less than one are exactly those with reduction zero; roots reducing to a specified nonzero  $c$  give the indicated shell direction. The multiplicities in both descriptions agree.  $\square$

The finite-degree initial-polynomial identity in the repository's polynomial report has thus acquired an analytic justification for infinite series: Weierstrass preparation makes the extra tail harmless *under an explicit Gauss convergence condition*. A formal series in  $K[[Z]]$  with no such condition is outside this theorem.

**Corollary 6.6** (Valuation Rouché theorem). *If  $f, g \in A_R$  and  $\|g - f\|_R < \|f\|_R$ , then  $f$  and  $g$  have the same closed-disk, open-disk, shell and residue-direction zero counts at radius  $R$ .*

*Proof.* The normalized reductions in (6.4) are identical, since the normalized difference has Gauss norm less than one. Apply Theorem 6.5.  $\square$

The conclusion protects counts, not labels, equality of roots, or equality of all the pairwise distances in a cluster. Finer root-matching assertions need separation hypotheses. Nor does this statement evaluate  $f$  and  $g$  on an ordinary complex circle and compare their ordinary complex magnitudes.

## 6.4 A simple residual root gives an actual analytic root

**Lemma 6.7** (Unit-derivative Hensel lifting). *Let  $f \in \mathcal{O}_K\langle U \rangle$ ,  $a \in \mathcal{O}_K$ , and suppose  $|f(a)|_v < 1$  and  $|f'(a)|_v = 1$ . There is exactly one root  $b$  of  $f$  in  $a + \mathfrak{m}_K$ . Newton iteration from  $a$  converges to  $b$ ; each new residual norm is at most the square of the preceding one.*

*Proof.* Taylor expansion about any point of  $\mathcal{O}_K$  has coefficients in  $\mathcal{O}_K$ ; this follows first for polynomials from the integral binomial coefficients, and then for Tate series by continuity. A displacement  $u$  of norm less than one changes the derivative by a term of norm less than one. The Newton correction  $-f(a)/f'(a)$  therefore has norm less than one, the next derivative remains a unit, and the Taylor remainder is bounded by the square of the correction. Iteration gives a Cauchy sequence, hence a root. If  $b_1, b_2$  are in the same residual disk, the divided difference  $(f(b_1) - f(b_2))/(b_1 - b_2)$  is a unit when  $b_1 \neq b_2$ ; its leading term is  $f'(a)$ . It cannot vanish, so two roots there are impossible.  $\square$

## 7 Berkovich disks: points that are not surcomplex numbers

### 7.1 The spectrum of a Banach algebra

For  $A_R$  define  $\mathcal{M}(A_R)$  to be the set of maps  $p : A_R \rightarrow \mathbb{R}_{\geq 0}$  satisfying

$$p(0) = 0, \quad p(1) = 1, \quad p(fg) = p(f)p(g), \quad p(f+g) \leq \max\{p(f), p(g)\},$$

which extend  $|\cdot|_v$  on  $K$  and obey  $p(f) \leq \|f\|_R$ . The topology is the weakest one in which every real-valued map  $p \mapsto p(f)$  is continuous. This is the Berkovich closed disk, with the usual bounded multiplicative-seminorm definition specialized to our algebra [2, Definition 2.2.2.1]. Using the bound with constant one loses no seminorms: if  $p(f) \leq C\|f\|_R$  for some fixed  $C$ , then multiplicativity gives  $p(f)^n = p(f^n) \leq C(\|f\|_R)^n$  for every positive integer  $n$ . Taking  $n$ th roots and letting  $n \rightarrow \infty$  gives  $p(f) \leq \|f\|_R$ .

A classical point  $z \in D_R(K)$  gives  $p_z(f) = |f(z)|_v$ . These are not all points. The Gauss norm itself is a multiplicative seminorm by Proposition 4.1, so it defines a point  $\zeta_{0,R}$ .

**Proposition 7.1.** *The space  $\mathcal{M}(A_R)$  is nonempty, compact and Hausdorff. Its Gauss point simultaneously realizes the Gauss norm of every function.*

*Proof.* Embed it in the product

$$\prod_{f \in A_R} [0, \|f\|_R].$$

This is a set-indexed product of compact Hausdorff intervals. The equations and inequalities defining a seminorm, and the equations on constants, are closed conditions on its coordinates. Hence  $\mathcal{M}(A_R)$  is a closed subspace of that compact Hausdorff product. It is nonempty because it contains evaluation at zero. At the Gauss point the coordinate  $p(f)$  is, by definition,  $\|f\|_R$ .  $\square$

A one-point boundary on which all spectral suprema are attained is thus available. In the terminology of affinoid algebras, the Gauss point is the Shilov boundary of the closed disk. No classical point can do this job for every function. For instance, on the unit disk the Gauss norm of  $Z - a$  is one for every  $a \in \mathcal{O}_K$ , while evaluation at  $a$  makes it zero. The new point  $\zeta_{0,1}$  is a seminorm, not an exotic surreal number waiting to be named.

### 7.2 Disk points and the simplification for the full Hahn field

For  $a \in K$  and  $r \geq 0$ , define a seminorm on  $K[Z]$  by

$$|P|_{a,r} = \max_n |b_n|_v r^n, \quad P(Z) = \sum_n b_n (Z - a)^n. \quad (7.1)$$

At  $r = 0$  this is evaluation at  $a$ . For  $r > 0$ , scale the coordinate by an element of norm  $r$  and use the reduction proof of multiplicativity. If  $|a|_v \leq R$  and  $r \leq R$ , it extends continuously to  $A_R$  and defines  $\zeta_{a,r} \in \mathcal{M}(A_R)$ .

The standard classification over an algebraically closed complete valued field has four types: classical points; disk points whose positive radius belongs to the value group; disk points with radius outside it; and points arising from nested disks with empty intersection. This classification is imported, not proved merely by drawing a tree [3, Lemma I.2.11, Proposition I.2.12 and Remark I.2.13].

**Corollary 7.2.** *Over  $K = \mathbb{C}((t^{\mathbb{R}}))$  the Berkovich affine line has only types 1 and 2. Every point is represented by some  $\zeta_{a,r}$ , with  $r \geq 0$ . Over  $\mathbb{C}((t^{\mathbb{Q}}))$  types 1, 2 and 3 occur, but type 4 still does not occur.*

*Proof.* For  $K$ , every positive real radius is in the value group, excluding type 3; spherical completeness excludes type 4. The coefficient-gluing proof of Proposition 2.2 also works over the divisible subgroup  $\mathbb{Q}$ , so type 4 is excluded there too. But a radius such as  $e^{-\sqrt{2}}$  is not in  $e^{-\mathbb{Q}}$ , and its disk seminorm is of type 3 over that field.  $\square$

The distinction between the full Hahn field and its smaller metric subfields is therefore geometrically visible. An assertion about the types over a completed Puiseux field must not be copied unchanged to a spherically complete Hahn field.

### 7.3 Where paths come from

For fixed  $a$ , the map  $r \mapsto \zeta_{a,r}$  is continuous. On polynomials this follows immediately from (7.1); on a fixed disk, uniform polynomial approximation gives the same conclusion for Tate functions. Moreover,

$$\zeta_{a,r} = \zeta_{b,r} \iff |a - b|_v \leq r \quad (r > 0). \quad (7.2)$$

One direction follows by recentering and the ultrametric inequality; for the other evaluate  $Z - a$  at the two seminorms.

Two distinct classical points  $a, b$  can consequently be joined by increasing the radius from zero to  $d = |a - b|_v$  about  $a$ , then decreasing it from  $d$  to zero about  $b$ . The two radius paths meet at their first common containing disk. This path uses nonclassical points. It is not a continuous ordinary-real path taking its values only in the ultrametric field  $K$ .

More generally for positive-radius disk points  $x = \zeta_{a,r}$  and  $y = \zeta_{b,s}$ , put

$$M = \max\{r, s, |a - b|_v\}.$$

Their common upper disk is  $\zeta_{a,M} = \zeta_{b,M}$ , and the logarithmic length of the intervening tree segment is

$$d_{\text{tree}}(x, y) = 2 \log M - \log r - \log s. \quad (7.3)$$

This is the path-length convention on nonclassical segments, not a claim that the entire compact weak topology is this metric. Classical endpoints have infinite logarithmic distance from a fixed positive-radius point even though the radius-parameterized path reaches them continuously. Unique arcwise connectedness is a standard theorem of the Berkovich line [3, Sections I.6 and I.8]; the formulas here describe those arcs in the particular field where every point is a disk point.

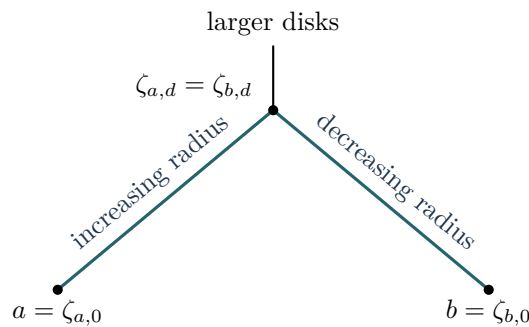


Figure 1: A schematic path through the first common disk, with  $d = |a - b|_v$ . Only the two selected branches and their common upper segment are shown; this is not an ordinary picture of points in a complex disk.

## 7.4 Residue directions replace the circle of boundary points

The Gauss point of the unit disk has one inward branch for each residue class  $c \in \mathbb{C}$ . The corresponding open disk is

$$D^-(c, 1) = \{x : |Z - c|_x < 1\}.$$

The Berkovich closed unit disk is the disjoint union of its Gauss point and these open disks. One can see the set decomposition directly in our disk model: a proper disk of radius less than one has a unique residue center; a disk of radius one with center in  $\mathcal{O}_K$  is the Gauss point itself. The standard tree topology makes these precisely the connected branches. In the projective line there is also the outward direction, giving the indexing set  $\mathbb{P}^1(\mathbb{C})$  [3, Sections I.3 and I.8].

This explains the initial polynomial in (6.4): its roots specify which residue branches contain zeros and with what multiplicities. The finite root geometry is no longer an isolated algebraic construction; it is the branch geometry of an analytic spectrum. Nevertheless its “boundary” is not a circle parametrized by an ordinary angle.

## 8 Norm profiles, integer slopes, and a Jensen formula

### 8.1 The Newton profile as a function on a radial segment

Suppose  $0 \neq f = \sum a_n Z^n$  is analytic on all disks in a nonempty interval of radii. For  $x = \log R$  put

$$L_f(x) = \log \|f\|_{e^x} = \max_{n: a_n \neq 0} \{-v(a_n) + nx\}. \quad (8.1)$$

It is also the logarithm of  $f$  evaluated at the radial Berkovich point  $\zeta_{0,e^x}$ .

**Proposition 8.1.** *On every compact subinterval of an open interval of analyticity,  $L_f$  is the maximum of finitely many affine functions with nonnegative integer slopes. It is therefore convex and piecewise affine. At a radius  $R = e^x$  where both one-sided neighborhoods are available,*

$$L'_f(x-) = m_R, \quad L'_f(x+) = d_R.$$

*Its slope jump is the number of zeros on the valuation shell of radius  $R$ , counted with multiplicity.*

*Proof.* At the largest radius in a compact interval the weighted coefficient norms tend to zero, and hence the far tail is uniformly small at every smaller radius in the interval. One fixed nonzero term has a positive minimum norm on that interval. Eventually every tail term is smaller than that minimum, so it cannot attain the maximum. Only finitely many affine functions remain.

At an intersection of finitely many active affine functions the left slope of their maximum is the smallest active slope and the right slope is the largest. These are exactly the indices in (6.3). The shell count is Theorem 6.5.  $\square$

At the outer boundary of a domain the algebraic numbers  $m_R, d_R$  still make sense even when a right derivative has no domain. This minor qualification prevents an implicit analytic continuation assumption.

### 8.2 A product formula on a disk

For a polynomial  $P = c \prod_j (Z - \alpha_j)$ , multiplicativity gives immediately

$$\|P\|_R = |c|_v \prod_j \max\{R, |\alpha_j|_v\}. \quad (8.2)$$

A nonpolynomial Tate function has a similar formula after preparation, its unit contributing a constant norm on the disk.

**Theorem 8.2** (Valuation Jensen formula). *If  $f \in A_R$  and  $f(0) \neq 0$ , and  $\alpha_1, \dots, \alpha_d$  are its zeros in  $D_R(K)$  with multiplicity, then*

$$\log \|f\|_R - \log |f(0)|_v = \sum_{j=1}^d \log \frac{R}{|\alpha_j|_v}. \quad (8.3)$$

*A zero on the boundary shell contributes zero to the displayed sum.*

*Proof.* Prepare  $f$  after scaling to the unit disk. The unit criterion gives  $\|u\|_R = |u(0)|_v$  for its unit factor. Apply (8.2) to the prepared polynomial; all its roots have norm at most  $R$ . Divide by the product formula at zero and take real logarithms.  $\square$

No angular average occurs in this proof. Formula (8.3) is a statement about the Gauss norm and root radii, and its differentiation along a radial tree segment gives the integer-slope count. It should not be identified with the order-modulus Jensen theorem in the repository's polynomial report: that theorem concerns a different geometry [8].

## 9 Differentiation, primitives, and a local differential equation

### 9.1 The coefficient field is held fixed

The differential in this article is

$$\frac{d}{dZ} \left( \sum a_n Z^n \right) = \sum_{n \geq 1} n a_n Z^{n-1}, \quad a_n \in K.$$

Thus  $dt/dZ = 0$ . It is not a derivation on **No** with a chosen value on  $\omega$ , and no compatibility with a surreal differential-field structure is asserted. This is the coordinate differential used for analytic functions over a fixed ground field.

**Proposition 9.1.** *For every  $R > 0$ , differentiation is a bounded map  $A_R \rightarrow A_R$ , and*

$$\|f'\|_R \leq R^{-1} \|f\|_R.$$

*The zero-constant primitive*

$$Jf = \sum_{n \geq 0} \frac{a_n}{n+1} Z^{n+1} \quad (9.1)$$

*belongs to  $A_R$ , satisfies  $(Jf)' = f$ , and obeys  $\|Jf\|_R = R \|f\|_R$ . Consequently the global analytic de Rham complex on the disk has*

$$H_{\text{dR}}^0(A_R) = K, \quad H_{\text{dR}}^1(A_R) = 0.$$

*Proof.* Every nonzero integer and its inverse have valuation norm one. The displayed bounds follow term by term from the Gauss norm, and the same estimates preserve the vanishing tails. The kernel of differentiation consists exactly of the constants, since the characteristic is zero, and (9.1) proves surjectivity. Here  $H_{\text{dR}}^1(A_R)$  means the quotient  $A_R dZ/dA_R$  of global analytic forms, which is all that is needed below.  $\square$

The norm-one denominator property is essential. In a field with nontrivial  $p$ -adic norm on the integers, division by  $n+1$  can destroy convergence at the boundary of a closed disk. Nothing in Proposition 9.1 is a blanket assertion for all complete non-Archimedean fields.

## 9.2 Why having primitives does not solve every equation globally

Consider  $y' = y$ ,  $y(0) = 1$ . Coefficient comparison forces the unique formal solution  $y = \sum Z^n/n!$ . It is not in  $A_1$ , although every individual function of  $A_1$  has a primitive in  $A_1$ . Solving a differential equation is not the same operation as taking one primitive; infinitely many successive integrations require their own convergence estimate.

There is a clean local replacement. Give matrices over  $A_R$  the maximum-entry Gauss norm, which is submultiplicative by the ultrametric inequality.

**Proposition 9.2.** *If a matrix  $B(Z)$  over  $A_R$  satisfies  $R\|B\|_R < 1$ , the initial-value problem*

$$Y' = BY, \quad Y(0) = I$$

*has a unique solution over  $A_R$ . It satisfies  $\|Y - I\|_R < 1$  and is an invertible matrix over  $A_R$ .*

*Proof.* The map  $Y \mapsto I + J(BY)$  is a contraction, with constant at most  $R\|B\|_R$ . Its fixed point is a solution. Differentiation and evaluation at zero show that any other solution is the same fixed point. Iterating from  $I$  gives  $\|Y\|_R \leq 1$  and  $\|Y - I\|_R \leq R\|B\|_R < 1$ ; the matrix geometric series in  $I - Y$  supplies the inverse.  $\square$

The condition is sufficient, not necessary. Applied after shrinking a disk on which  $B$  is analytic, it gives a genuinely local existence theorem. It neither produces a fundamental matrix on every closed disk nor supports a universal surreal exponential at arbitrary scales.

## 10 Analytic annuli, residues, and the failure of a naive comparison

### 10.1 Laurent series on a closed annulus

Fix  $0 < r < R$ . Define

$$A[r, R] = \left\{ \sum_{n \in \mathbb{Z}} a_n Z^n : \begin{array}{ll} |a_n|_v R^n \rightarrow 0 & (n \rightarrow +\infty), \\ |a_n|_v r^n \rightarrow 0 & (n \rightarrow -\infty) \end{array} \right\}. \quad (10.1)$$

Its norm is

$$\|f\|_{[r, R]} = \max \left\{ \sup_n |a_n|_v r^n, \sup_n |a_n|_v R^n \right\}.$$

This is a complete submultiplicative norm: the algebra is the completion of Laurent polynomials for the maximum of their two endpoint Gauss norms. It is *not generally multiplicative*. Indeed the norms of  $Z$  and  $Z^{-1}$  are  $R$  and  $r^{-1}$ , while their product is one. Evaluation converges on  $r \leq |z|_v \leq R$ ; the two endpoint estimates justify the Cauchy product.

If  $a, b \in K^*$  have norms  $R, r$ , the coordinates  $U = Z/a$  and  $V = b/Z$  satisfy  $UV = b/a$ . Thus (10.1) is the usual annulus affinoid algebra, also written  $K\langle R^{-1}Z, rZ^{-1} \rangle$ , with the familiar presentation  $K\langle U, V \rangle / (UV - b/a)$  [2, Section 3].

The Laurent coefficients are intrinsic to the analytic function, not just to a chosen notation. To see injectivity of evaluation, take a nonzero Laurent series and a radius  $\rho \in [r, R]$ . After scaling and normalization its reduction is a nonzero Laurent polynomial. Choose a nonzero complex residue which is not a root; evaluation at the corresponding point of norm  $\rho$  is nonzero. This is the same finite initial-polynomial test used on a disk.

## 10.2 The residue is the complete primitive obstruction

**Definition 10.1.** For an analytic one-form on the annulus define

$$\mathrm{Res}_{[r,R]} \left( \sum_{n \in \mathbb{Z}} a_n Z^n dZ \right) = a_{-1}.$$

The coordinate  $Z$  includes an orientation of the annulus: reversing it by  $Z = W^{-1}$  reverses the residue.

**Theorem 10.2.** *An analytic one-form  $f(Z) dZ$  on  $A[r, R]$  has an analytic primitive on that same closed annulus if and only if its residue is zero. More precisely,*

$$0 \longrightarrow K \longrightarrow A[r, R] \xrightarrow{d} A[r, R] dZ \xrightarrow{\mathrm{Res}_{[r,R]}} K \longrightarrow 0 \quad (10.2)$$

is exact. In particular  $H_{\mathrm{dR}}^1(A[r, R]) = K[dZ/Z]$ .

*Proof.* A derivative has no  $Z^{-1}$  coefficient, so a primitive forces zero residue. Conversely, when  $a_{-1} = 0$  set

$$P(Z) = \sum_{n \neq -1} \frac{a_n}{n+1} Z^{n+1}.$$

At each endpoint  $\rho = r, R$  the corresponding term norm is  $\rho |a_n|_v \rho^n$ , since  $|n+1|_v = 1$ . Both tails still tend to zero, so  $P$  belongs to the same annulus algebra and  $P' = f$ . Its norm is at most  $R \|f\|_{[r,R]}$ . Constants are the derivative kernel; the residue is onto because  $c dZ/Z$  has residue  $c$ .  $\square$

The repository already discusses the formal residue quotient  $K((Z)) dZ / dK((Z))$  [9]. The point here is an analytic statement with *both* Laurent tails present and convergence preserved on the very same closed annulus. No shrinking is hidden in the primitive formula.

**Proposition 10.3.** *Under a monomial covering  $Z = cW^m$ ,  $c \neq 0$ ,  $m \in \mathbb{N}_{>0}$ , choose the source annulus so that  $r \leq |c|_v |W|_v^m \leq R$ . Then*

$$\mathrm{Res}((cW^m)^*(f(Z) dZ)) = m \mathrm{Res}(f(Z) dZ).$$

Also, if  $\|u\|_{[r,R]} < 1$ , the Banach series  $\log(1+u) = \sum_{k \geq 1} (-1)^{k+1} u^k / k$  converges, and  $\mathrm{Res}(d \log(1+u)) = 0$ .

*Proof.* The pullback of a monomial is  $a_n m c^{n+1} W^{m(n+1)-1} dW$ ; exponent  $-1$  occurs precisely at  $n = -1$ . The annulus endpoint conditions justify passing through the convergent series. For the logarithm,  $|k|_v = 1$  and the term norms are bounded by  $\|u\|_{[r,R]}^k \rightarrow 0$ . Differentiation is bounded on the annulus, so it commutes with the norm limit. Exactness in (10.2) then gives zero residue.  $\square$

## 10.3 The Berkovich annulus is contractible, nevertheless

Let  $X[r, R]$  be the Berkovich annulus

$$\{x : r \leq |Z|_x \leq R\}.$$

Its skeleton is the interval of disk points

$$\Sigma[r, R] = \{\zeta_{0,\rho} : r \leq \rho \leq R\}. \quad (10.3)$$

The general deformation-retraction theorem is standard: the open-annulus construction in [3, Lemma I.7.9 and Proposition I.7.11] restricts to the inverse image of a closed skeleton interval, giving a closed annulus. For our spherically complete algebraically closed field it has an explicit proof worth recording.

**Proposition 10.4.** *The space  $X[r, R]$  strongly deformation retracts onto  $\Sigma[r, R]$  and is therefore contractible.*

*Proof.* By Corollary 7.2, each point can be written as  $x = \zeta_{a,s}$ . Choose the representation with  $s \leq |a|_v$ ; a skeleton point of radius  $s$  can be recentered at any  $a$  of norm  $s$ . Then  $\rho = |Z|_x = |a|_v \in [r, R]$ . For  $0 \leq \tau \leq 1$  define

$$H(\tau, x) = \zeta_{a, \max\{s, \tau\rho\}}.$$

Changing a representing center by distance at most  $s$  does not change any of the enlarged disks, so the map is well defined. It stays in the annulus, starts at  $x$ , and ends at  $\zeta_{a,\rho} = \zeta_{0,\rho}$ . Skeleton points remain fixed.

For joint continuity it suffices first to test linear polynomials:

$$|Z - b|_{H(\tau, x)} = \max\{|Z - b|_x, \tau|Z|_x\}.$$

This is continuous in  $(\tau, x)$ . Every polynomial factors into linear terms because  $K$  is algebraically closed, so multiplicativity gives continuity for all polynomials. Laurent polynomials follow by division by powers of  $Z$ , whose norm remains bounded below by  $r$ . Every annulus function is a uniform limit of Laurent polynomials, and all these seminorms are bounded by the annulus norm. Continuity follows for the defining coordinates of the spectrum. Finally  $\rho \mapsto \zeta_{0,\rho}$  identifies the skeleton with  $[r, R]$ , with continuous inverse  $x \mapsto |Z|_x$ . The interval is contractible.  $\square$

Combining this proposition with Theorem 10.2 gives

$$H_{\text{sing}}^1(X[r, R]; K) = 0, \quad H_{\text{dR}}^1(A[r, R]) \simeq K. \quad (10.4)$$

There is no contradiction. The analytic differential forms over  $K$  used here are not the real differential forms on the underlying tree-like topological space. An ordinary complex de Rham comparison cannot be assumed in this new category. In particular  $dZ/Z$  has a nonzero analytic residue although there is no topological loop around a hole in this Berkovich annulus. This is a concrete reason not to regard the skeleton as an ordinary circular contour.

#### 10.4 An argument principle with no hidden contour

Suppose a rational function  $f \in K(Z)^*$  has no zero or pole on the closed annulus. Factoring it gives

$$\frac{f'}{f} = \sum_j \frac{m_j}{Z - \alpha_j} - \sum_k \frac{n_k}{Z - \beta_k}.$$

A root or pole has norm either less than  $r$  or greater than  $R$ . If  $|\alpha|_v < r$ , then

$$(Z - \alpha)^{-1} = \sum_{k \geq 0} \alpha^k Z^{-k-1}$$

converges in the annulus algebra and has residue one. If  $|\alpha|_v > R$ , expanding in  $Z/\alpha$  gives no negative power and residue zero. Consequently

$$\text{Res}_{[r, R]} \left( \frac{f'}{f} dZ \right) = \#\{\text{zeros in } |z|_v < r\} - \#\{\text{poles in } |z|_v < r\}, \quad (10.5)$$

with multiplicities. This is an exact Laurent-series argument principle; it has not been defined as a Riemann integral along a  $K$ -valued circle.

For a Tate function analytic on a larger disk and nonvanishing on the annulus, preparation gives the analogous formula. The polynomial factor contributes its inner zeros; the disk-analytic unit has logarithmic derivative analytic on the full disk and hence zero annular residue. This version will apply to the nonpolynomial example in Section 12.

## 10.5 Actual, annular, and cluster residues

Let  $\varepsilon \in \mathfrak{m}_K \setminus \{0\}$ . The rational form  $dZ/(Z - \varepsilon)$  has local residue zero at the actual point  $Z = 0$  and local residue one at  $Z = \varepsilon$ . On any annulus with  $|\varepsilon|_v < r < R$  its Laurent residue is one. The Hahn expansion

$$\frac{1}{Z - \varepsilon} = \sum_{n \geq 0} \varepsilon^n Z^{-n-1}$$

also has coefficientwise cluster residue one at the ordinary center zero, in the overlapping rational framework of the repository. These answers agree where their definitions concern the same cluster, and differ where they concern different actual points. There is no single symbol “residue at zero” that may safely replace all three constructions.

## 11 Entire functions and their canonical products

### 11.1 Entireness means every real valuation radius

Define

$$\mathcal{E}(K) = \bigcap_{R > 0} A_R = \left\{ \sum_{n \geq 0} a_n Z^n : v(a_n) + n\beta \rightarrow +\infty \text{ for every } \beta \in \mathbb{R} \right\}. \quad (11.1)$$

The intersection is taken inside  $K[[Z]]$ . Elements are analytic at every point of this rank-one field. This is not the assertion that the same defining sum can be evaluated at every point of  $\mathbf{No}[i]$ .

**Theorem 11.1** (Growth and zero-free rigidity). *Let  $f \in \mathcal{E}(K)$ .*

1. *If  $|f(z)|_v \leq M$  for every  $z \in K$  and some real  $M$ , then  $f$  is constant.*
2. *More generally, if  $\|f\|_R \leq CR^d$  for all sufficiently large  $R$ , where  $d \in \mathbb{N}$ , then  $f$  is a polynomial of degree at most  $d$ .*
3. *If  $f$  is zero-free on  $K$ , then  $f$  is a nonzero constant.*

*Proof.* By Proposition 4.2, the first hypothesis bounds every Gauss norm by  $M$ . Thus  $|a_n|_v R^n \leq M$  for all  $R$ , forcing  $a_n = 0$  for  $n > 0$ . For the second assertion the same coefficient bound reads  $|a_n|_v \leq CR^{d-n}$ , and letting  $R \rightarrow \infty$  eliminates  $n > d$ .

For the third assertion, root counting gives  $d_R = 0$  at every radius. In particular  $a_0 \neq 0$  and its term dominates all higher terms at every radius. A nonzero  $a_n$  with  $n > 0$  would satisfy  $|a_n|_v R^n > |a_0|_v$  at some sufficiently large radius, a contradiction.  $\square$

Zero-free entire rigidity is stronger than bounded-entire rigidity here. It is not a claim about ordinary complex entire functions: the usual exponential is excluded from (11.1). The following product theorem also shows that a nonpolynomial valuation-entire function must have infinitely many zeros escaping to infinity in valuation norm.

### 11.2 An infinite product without exponential correction factors

**Theorem 11.2.** *Let  $0 \neq f \in \mathcal{E}(K)$  have a zero of order  $m$  at zero, and let  $(\alpha_j)$  be its nonzero zeros, repeated with their multiplicities. There are finitely many in each bounded valuation disk, so in the infinite case they can be enumerated with  $|\alpha_j|_v \rightarrow \infty$ . Then*

$$f(Z) = cZ^m \prod_j \left(1 - \frac{Z}{\alpha_j}\right), \quad c = [Z^m]f. \quad (11.2)$$

*The product converges in  $A_R$  for every  $R > 0$ , and its limit is independent of the escaping enumeration. A nonzero entire function with only finitely many zeros is a polynomial.*

*Proof.* Theorem 6.5 gives finiteness on each disk. The disks of radii  $e^n$ ,  $n \geq 0$ , cover  $K$ , so the zero multiset is countable and can be enumerated as stated.

Fix  $R$ . Apart from finitely many factors,  $R/|\alpha_j|_v < 1$  and tends to zero. For elements  $u_j$  of norm less than one, finite multiplication and the ultrametric inequality give

$$\left\| \prod_{j=p}^q (1 + u_j) - 1 \right\| \leq \max_{p \leq j \leq q} \|u_j\|.$$

Apply this in  $A_R$  with  $u_j = -Z/\alpha_j$ . The tails converge to units; indeed a sufficiently far tail differs from one by norm less than one, so Corollary 6.3, after rescaling, gives its inverse. The finite initial product causes no difficulty. For each positive tolerance there are only finitely many factors whose errors are at least that large. The tail estimate therefore shows that every enumeration has the same limit, once the finitely many exceptional factors for the required precision have been included. The limits on different disks agree as formal power series. Denote this product by  $P$  and set  $G = Z^m P$ . On a fixed disk, split  $P$  into the finitely many factors whose zeros lie in that disk and the product of the remaining factors. The latter is a unit: its factor errors have norms strictly less than one and tend to zero, so their supremum is strictly less than one. Thus  $G$  has exactly the zeros of  $f$ , with the same multiplicities, including order  $m$  at zero.

Both  $f/Z^m$  and  $G/Z^m = P$  have nonzero constant term, so  $Q = (f/Z^m)/P$  is defined in  $K[[Z]]$ . On each disk, prepare the rescaled functions  $f$  and  $G$ . Their monic prepared polynomials agree because they have the same roots with multiplicity; cancelling them leaves a quotient of analytic units. Its Taylor series is  $Q$ , by uniqueness of formal division after cancelling  $Z^m$ . Hence  $Q \in A_R$  and is zero-free there for every  $R$ . Theorem 11.1 makes  $Q$  constant. Since  $P(0) = 1$ , its value is  $[Z^m]f$ , proving (11.2). A finite product is a polynomial.  $\square$

The product theorem needs no analog of the complex Weierstrass exponential factors. In a non-Archimedean Banach algebra, smallness of the individual tail factors already controls their whole product by a maximum, rather than an ordinary sum. This proof does not assert that all individual root coefficients can be computed effectively from an arbitrary Hahn presentation.

## 12 Case study: the entire series with quadratic Hahn exponents

### 12.1 The function, its antecedents, and its domain

Consider

$$F(Z) = \sum_{n \geq 0} t^{n^2} Z^n. \quad (12.1)$$

The plus-sign quadratic-exponent series already occurs in the repository's foundations report as a counterexample to transporting full-surcomplex all-scale rigidity to a fixed Hahn workspace [10]. It is also a partial theta function:

$$F(Z) = \Theta_0(tZ, t^2), \quad \Theta_0(x, q) = \sum_{n \geq 0} x^n q^{n(n-1)/2}.$$

Sokal studies the leading root of this classical partial theta family [4]. We do not claim to introduce the series or its first root. Our task is to work out its *whole* zero geometry in the present rank-one analytic category and connect that geometry to the preceding theorems.

Since  $n^2 + n\beta \rightarrow +\infty$  for every real  $\beta$ , we have  $F \in \mathcal{E}(K)$ . The same holds over  $\mathbb{C}((t^{\mathbb{Q}}))$  at every radius of that field, and  $F$  is nonpolynomial. Coefficient shifting gives the exact entire functional equation

$$F(Z) = 1 + tZF(t^2Z). \quad (12.2)$$

There is no limit  $t \rightarrow 0$  in these statements:  $t$  is the fixed Hahn monomial. The auxiliary notation  $q = t^2$  used below records a valuation grading.

## 12.2 The complete Newton profile

Writing  $x = \log R$ , formula (8.1) becomes

$$L_F(x) = \max_{n \geq 0} (nx - n^2). \quad (12.3)$$

The switch from  $n = m - 1$  to  $n = m$  occurs precisely at  $x = 2m - 1$ . Between  $2m - 1$  and  $2m + 1$  the unique active index is  $m$ . Hence for  $x \geq 0$ ,

$$L_F(x) = \frac{x^2}{4} - \text{dist}\left(\frac{x}{2}, \mathbb{N}\right)^2, \quad \frac{x^2}{4} - \frac{1}{4} \leq L_F(x) \leq \frac{x^2}{4}, \quad (12.4)$$

where  $\mathbb{N}$  includes zero. For  $x \leq 1$ , the norm profile is zero.

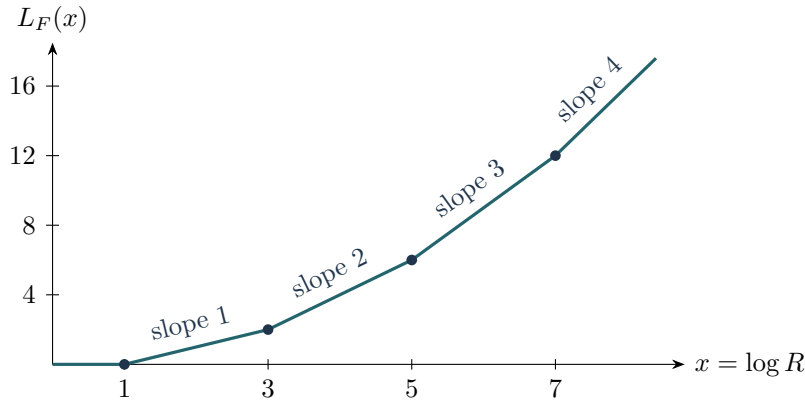


Figure 2: The exact radial norm profile of  $F$ . Each unit slope jump at an odd positive integer records one zero on the corresponding valuation shell. The axes are ordinary real logarithms of norms, not surreal coordinates.

**Theorem 12.1** (All zeros and their first terms). *For each integer  $m \geq 1$  there is exactly one zero  $z_m$  of  $F$  with*

$$v(z_m) = -(2m - 1). \quad (12.5)$$

*These are all the zeros of  $F$  in  $K$ . They are simple and belong to the negative part of the real-coefficient Hahn field. Writing*

$$z_m = t^{-(2m-1)} u_m,$$

*one has  $u_m \equiv -1 \pmod{\mathfrak{m}_K}$  and, more precisely,*

$$u_m = -1 + (-1)^m t^{m(m+1)} + O(t^{m(m+1)+2}), \quad (12.6)$$

$$z_m = -t^{-(2m-1)} + (-1)^m t^{m^2-m+1} + O(t^{m^2-m+3}). \quad (12.7)$$

*In fact  $u_m \in \mathbb{Z}[[t^2]] \subset K$ . Here  $O(t^a)$  means a Hahn series of valuation at least  $a$ , not an ordinary numerical estimate obtained by substituting a small real value for  $t$ .*

*Proof.* At  $R = e^{2m-1}$  the minimal coefficient valuations after rescaling occur precisely at indices  $m - 1, m$ . Normalize by setting

$$F_m(U) = t^{m(m-1)} F(t^{-(2m-1)} U) = \sum_{n \geq 0} t^{(n-m)(n-m+1)} U^n. \quad (12.8)$$

This is in  $\mathcal{O}_K\langle U \rangle$ , and its reduction is

$$\text{red}(F_m) = U^{m-1}(1 + U).$$

Thus there are exactly  $m - 1$  roots in the open unit disk and exactly one on its unit shell, counted with multiplicity. The shell root has reduction  $-1$  and is simple, either by the count or by Lemma 6.7. Rescaling gives (12.5).

The disk of radius  $e^{2m}$  has exactly  $m$  zeros, and already contains  $z_1, \dots, z_m$ . These disks exhaust  $K$ , so there are no additional zeros. Conjugation preserves  $F$  and the shell and fixes its unique root; hence the root has real Hahn coefficients. Its leading coefficient is negative, making it a negative element in the ordered real Hahn field.

For the sharper correction work near the unit  $U = -1$  and divide by the unit power  $U^{m-1}$ :

$$H_m(U) = U^{1-m}F_m(U) = \sum_{j \geq 1-m} t^{j(j-1)}U^j. \quad (12.9)$$

There are only finitely many negative powers. The two leading terms are  $1 + U$ , so  $H'_m(-1) = 1 + O(t^2)$ . At  $U = -1$ , pair the terms with indices  $j$  and  $1 - j$ . Their powers of  $t$  agree and their signs are opposite. All indices from  $1 - m$  through  $m$  cancel, leaving

$$H_m(-1) = \sum_{j \geq m+1} (-1)^j t^{j(j-1)} = (-1)^{m+1} t^{m(m+1)} + O(t^{m(m+1)+2}).$$

The use of the Laurent function  $H_m$  is local to  $|U + 1|_v < 1$ : each negative power of  $U = -1 + V$  has a binomial expansion with integral coefficients converging for  $|V|_v < 1$ . The Taylor coefficients of  $F_m(-1 + V)$  are in  $\mathcal{O}_K$ , so the same holds for those of  $H_m(-1 + V)$ . Thus its Taylor remainder after the linear term is bounded by  $|V|_v^2$ . Writing  $\delta = u_m + 1 \in \mathfrak{m}_K$  and  $A = m(m + 1)$ , the equation  $H_m(-1 + \delta) = 0$  has a unit linear coefficient and a quadratic remainder of strictly smaller norm than  $\delta$ . Hence  $v(\delta) = v(H_m(-1)) = A$ , and then

$$\delta = -H_m(-1)/H'_m(-1) + O(t^{2A}).$$

The reciprocal of  $H'_m(-1) = 1 + O(t^2)$  differs from one by  $O(t^2)$ , and  $2A \geq A + 2$ . This proves (12.6); multiplying by  $t^{-(2m-1)}$  gives (12.7). No value of  $U^{-1}$  at zero, or analyticity of  $H_m$  on the entire closed unit disk, was used.

Finally use  $q = t^2$  in (12.8). Every exponent  $(n - m)(n - m + 1)/2$  is a nonnegative integer, and at any fixed  $q$ -degree only finitely many powers of  $U$  occur. Solving coefficient by coefficient for  $U = -1 + \sum_{k \geq 1} c_k q^k$ , the coefficient of the new unknown is always the nonzero derivative of  $U^{m-1}(1 + U)$  at  $-1$ , namely  $(-1)^{m-1}$ . Induction gives  $c_k \in \mathbb{Z}$ . The resulting  $q$ -adic series converges in  $K$  and is the same unique root. Thus  $u_m \in \mathbb{Z}[[t^2]]$ .  $\square$

### 12.3 Low-order expansions and a simple-root residue

The accompanying exact program obtains the following expansions, where  $q = t^2$ :

$$\begin{aligned} u_1 &= -1 - q - 2q^2 - 4q^3 - 9q^4 - 21q^5 - 52q^6 + O(q^7), \\ u_2 &= -1 + q^3 + 3q^4 + 9q^5 + O(q^6), \\ u_3 &= -1 - q^6 - 3q^7 - 9q^8 + O(q^9), \\ u_4 &= -1 + q^{10} + 3q^{11} + 9q^{12} + O(q^{13}). \end{aligned}$$

The first row agrees with the negative of the series  $\xi_0(q)$  in Sokal's Eq. (1.7) [4]; the agreement is a check of normalization, not independent evidence for novelty. The growing first-correction exponent  $m(m + 1)$  is the important scale-separation fact here.

**Corollary 12.2.** *At the simple zero  $z_m$ ,*

$$v(F'(z_m)) = -m^2 + 3m - 1, \quad \text{lc}(F'(z_m)) = (-1)^{m-1}. \quad (12.10)$$

*Consequently the local residue of  $dZ/F(Z)$  at  $z_m$  has valuation  $m^2 - 3m + 1$  and leading coefficient  $(-1)^{m-1}$ .*

*Proof.* Differentiating (12.8) gives

$$F'_m(u_m) = t^{m(m-1)-(2m-1)} F'(z_m).$$

The left side has valuation zero and reduction  $(U^{m-1}(1+U))'_{U=-1} = (-1)^{m-1}$ . This yields (12.10). The residue at a simple zero is  $1/F'(z_m)$ , which reverses the valuation and inverts the leading coefficient.  $\square$

## 12.4 The global product and coefficient identities

Theorem 11.2 now gives the complete product

$$F(Z) = \prod_{m \geq 1} \left(1 - \frac{Z}{z_m}\right). \quad (12.11)$$

It converges in the Gauss norm on every fixed disk. Indeed  $|z_m^{-1}|_v = e^{-(2m-1)}$ , so each tail factor approaches one uniformly there. There is no extra zero-free entire factor to determine, and  $F(0) = 1$  fixes the constant.

Comparing coefficients gives an infinite family of exact identities. For  $k \geq 1$ ,

$$\sum_{1 \leq i_1 < \dots < i_k} \frac{1}{z_{i_1} \cdots z_{i_k}} = (-1)^k t^{k^2}. \quad (12.12)$$

These sums genuinely converge: below any valuation cutoff only finitely many index tuples can occur, since  $v(z_j^{-1}) = 2j - 1$ . For  $k = 1$  the identity is  $\sum_{m \geq 1} z_m^{-1} = -t$ . The smallest possible valuation on the left of (12.12) is  $1 + 3 + \dots + (2k - 1) = k^2$ , as the exact right side also shows.

If an annulus lies strictly between two successive zero shells,

$$e^{2m-1} < r < R < e^{2m+1},$$

then the analytic argument principle of Section 10 yields

$$\text{Res}_{[r,R]} \left( \frac{F'}{F} dZ \right) = m. \quad (12.13)$$

It measures the first  $m$  zeros inside the inner radius, not a number of ordinary turns made by a path. The same integer is the slope of the radial norm profile throughout that interval.

## 12.5 Why the series is not an all-surcomplex entire function

At the surcomplex argument  $Z = t^{-\omega}$ , the  $n$ th term of (12.1) would have valuation

$$n^2 - n\omega.$$

These exponents strictly decrease with  $n$ , because consecutive differences are  $2n + 1 - \omega < 0$ . They form a set which is not well ordered. Thus the defining family is not strongly summable at this argument. It also does not converge in the full fine topology.

This does not rule out an unrelated or category-dependent extension of the function. It proves the precise needed statement: the same series which is entire over our rank-one field is not globally evaluable by its defining Hahn sum on all of  $\mathbf{No}[i]$ . The obstruction agrees with, rather than refutes, the repository's distinction between internal all-scale behavior and the proper-class all-scale rigidity framework [10].

## 13 Certified computation and the executed finite checks

### 13.1 The information an analytic representation must carry

A symbolic representation of a Tate function needs more than a generator for its coefficients. It should carry a field or coefficient domain, a coordinate and radius, and a *tail certificate*. For example, a certificate can be a function  $B(N)$  with

$$\inf_{n \geq N} (v(a_n) + n\beta) \geq B(N), \quad B(N) \longrightarrow +\infty.$$

Then truncation after degree  $N$  has Gauss error at most  $e^{-B(N)}$  on  $R = e^{-\beta}$ . If that error is strictly below the norm of the retained polynomial, Corollary 6.6 certifies the same initial polynomial and zero counts for the infinite function.

Such a certificate concerns the *function-degree tail*. Each retained coefficient can itself be an infinite Hahn series, whose exact comparison or leading-term access requires a separate representation contract. This is consistent with the capability distinctions in the repository's computer-algebra report [12]. Neither Banach completeness nor a support theorem is a universal finite-time algorithm for arbitrary set-theoretic input.

### 13.2 An exact tail certificate for the quadratic example

For  $F_N(Z) = \sum_{n=0}^N t^{n^2} Z^n$  and  $R = e^{-\beta}$ , if  $N + 1 \geq -\beta/2$ , then

$$\|F - F_N\|_R = \exp\left(-((N + 1)^2 + (N + 1)\beta)\right). \quad (13.1)$$

Indeed the quadratic  $n^2 + n\beta$  is increasing for integers  $n \geq N + 1$ , so its first omitted value is the minimum tail valuation. This is an exact Gauss norm, not just a heuristic error estimate.

At the intermediate radius  $R = e^{2m}$ , the degree- $m$  truncation has its unique minimal weighted valuation  $-m^2$  at degree  $m$ , while the omitted tail starts at  $-m^2 + 1$ . At the shell radius  $R = e^{2m-1}$ , its minimal weighted valuation is  $-m(m - 1)$  and the omitted tail starts two valuation units higher. Thus degree  $m$  already certifies the disk and shell counts used in Theorem 12.1. Additional terms are needed for additional root coefficients, not to establish those counts.

### 13.3 What was actually run

The archive includes `code/verify-examples.py`, using only Python's standard library and exact `fractions.Fraction` arithmetic. It was executed under Python 3.13.5. The preserved report `data/verification.json` records every individual assertion and the computed root expansions; `data/verification-summary.txt` contains the human-readable run summary.

For each  $1 \leq m \leq 8$ , the program forms

$$G_m(U, q) = \sum_{n \geq 0} q^{(n-m)(n-m+1)/2} U^n \pmod{q^{N_m}}, \quad N_m = \frac{m(m+1)}{2} + 8.$$

Only finitely many  $n$  contribute at that precision. Formal Newton iteration in  $\mathbb{Q}[q]/(q^{N_m})$  starts at  $U = -1$ ; its derivative has invertible constant coefficient. The program checks the final residual exactly, the initial constant, the vanishing of all premature correction terms, the first correction sign, and the leading derivative coefficient.

| Finite check family                                        | Assertions | Passed |
|------------------------------------------------------------|------------|--------|
| Eight formal root expansions, five assertions each         | 40         | 40     |
| Displayed first-root coefficients through $q^6$            | 1          | 1      |
| Active Newton indices at forty corners and forty intervals | 80         | 80     |
| Quadratic-tail formula on a rational test grid             | 246        | 246    |
| Monomial residue-pullback exponents                        | 175        | 175    |
| Total for this one program                                 | 542        | 542    |

The report is deliberately a finite-check report. It does not verify arbitrary Hahn supports, infinite product convergence, Banach-space arguments, algebraic closedness, the classification of Berkovich points, or de Rham cohomology in a proof assistant. The infinite statements rely on the mathematical proofs in this article and the explicitly cited standard inputs, not on extrapolation from 542 successful tests.

## 14 Conclusions, boundaries, and natural continuations

### 14.1 What has been added to the documentation

The main addition is a usable interface between two kinds of mathematics. The repository's support-controlled Hahn coefficient families can be restricted to small valuation disks; in rank one this restriction carries an explicit Banach norm and genuine convergence. Tate algebras, in turn, sit strictly inside the polynomial-coefficient Hahn ring, rather than being a different notation for it. The ring chain (5.2) and the positive-rescaling map make both facts precise.

Once this interface is in place, preparation turns infinite series into finite zero geometry on every closed disk. The corresponding Berkovich points explain why the Gauss norm, residue directions and integer Newton slopes fit together. Analytic annuli supply a residue obstruction to primitives, but their contractible spectra demonstrate why ordinary complex topological intuitions must not be carried along without checking the differential-form category. Finally the entire partial-theta example provides a single calculation in which all these pieces interact: exact root shells, simple lifts, real ordering, residues, a canonical product, slope counts and certified truncations.

### 14.2 What is deliberately not being concluded

No topology or analytic spectrum has been constructed on the whole proper class  $\mathbf{No}[i]$ . No claim is made that an arbitrary set of surcomplex data is contained in a rank-one workspace preserving its natural valuation. No coefficientwise ordinary complex integral has been identified with a path integral in the Berkovich spectrum. No arbitrary-radius complex exponential, global differential field, or universal decision algorithm is supplied.

The class of entire functions here is exactly (11.1); it is not a synonym for every fine-local analytic class or for every coherent Hahn atlas in the repository. The derivative fixes  $K$ . All polynomial degrees are ordinary finite integers, all supports are sets, and all norm radii are positive ordinary real numbers. These qualifications are mathematical hypotheses, not merely implementation limitations.

### 14.3 Three directions that now have a precise starting point

A first continuation is several-variable Tate geometry after the explicit rescaling map, with finite analytic maps and residues compared to the existing multivariate Hahn-germ results. The starting ring must remain named: proving a theorem in  $K\langle Z_1, \dots, Z_n \rangle$  does not establish it unchanged in a radius-free Hahn-germ ring.

A second continuation is analytic curves with nontrivial skeleton graphs. The annulus calculated here has interval skeleton and is contractible, so it is a useful control example before studying genuine graph cycles. That extension would use more of the standard Berkovich curve theory rather than reinterpret an annulus as a circle.

A third continuation is a certified symbolic backend for rank-one analytic objects, beginning with integer or rational exponent coefficient streams and proven quadratic or linear tail bounds. It can separate root-count certificates from root-coefficient approximations and preserve the workspace and topology as part of an object's type. The finite program supplied here is a regression test for that future backend, not the backend itself.

## A Notation and category ledger

| Notation                                | Meaning                                                                                                                             |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{SC} = \mathbf{No}[i]$          | The proper class of surcomplex numbers; not our Banach field.                                                                       |
| $t^\gamma = \omega^{-\gamma}$           | Conway normal-form monomial, not an arbitrary surreal exponential convention.                                                       |
| $K = \mathbb{C}((t^{\mathbb{R}}))$      | The fixed set-sized full Hahn field, algebraically closed and spherically complete.                                                 |
| $v(a),  a _v = e^{-v(a)}$               | Leading real exponent and its real-valued multiplicative norm; every nonzero complex constant has norm one.                         |
| $R = e^{-\beta}$                        | An ordinary real norm radius; scaling is performed by the coefficient $t^\beta$ , not by treating $R$ as a norm-valued coefficient. |
| $A_R$                                   | Tate algebra on the closed valuation disk of radius $R$ .                                                                           |
| $\mathcal{P}, \mathcal{A}, \mathcal{F}$ | Polynomial-, convergent-germ-, and formal-coefficient Hahn families, each with a common well-ordered exponent support.              |
| $\mathcal{H}(D), \mathcal{R}_1$         | Fixed ordinary disk coefficient families, and their common-domain germ ring.                                                        |
| $\mathcal{E}^-$                         | The analytic open-unit-disk power-series ring; no common Hahn support is required across coefficients.                              |
| $\mathcal{E}(K)$                        | Functions analytic on every centered closed disk of finite real norm radius.                                                        |
| $\mathcal{M}(A_R), \zeta_{a,r}$         | A set of bounded multiplicative seminorms and a disk seminorm, not a field element or an additional surreal number.                 |
| $A[r, R]$                               | Analytic Laurent series with decay at both ends of the annulus.                                                                     |
| $d, H_{\mathrm{dR}}^1$                  | Coordinate differentiation fixing $K$ , and the cohomology of the displayed global analytic differential complex.                   |
| $O(t^a)$                                | A Hahn error of valuation at least $a$ , not a numerical specialization estimate.                                                   |
| $q = t^2$                               | The formal grading used to compute the normalized roots $u_m$ .                                                                     |

## B Reproduction and placement in the repository

The LaTeX source is standalone: its bibliography and diagrams are embedded, and no external image, bibliography database, font file, or repository checkout is needed. Build it with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Run the finite checks from the package directory with a separate output path:

```
python code/verify-examples.py -output /tmp/berkovich-verification.json
```

The maintained article is now placed in

```
docs/surcomplex/rank-one-berkovich/
```

The accompanying `repository-audit.md` records the original inspected paths and snapshot; `build-report.md` records the delivered build. Those historical records and the original `code/` and `data/` files are preserved. The current `README.md` records the maintained proof review and rebuilt PDF.

## References

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- [4] A. D. Sokal, *The leading root of the partial theta function*, Advances in Mathematics **229** (2012), 2603–2621. doi:10.1016/j.aim.2012.01.012; Preprint version 1 (2011), especially Eq. (1.7): [arXiv:1106.1003v1](#).
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- [6] *Surcomplex Analysis: Germs, coherent Hahn sections, and infinitesimal zero geometry*, research report in the Surreal repository, same snapshot. [LaTeX source](#); [report README](#). AI-assisted, unrefereed repository draft.
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- [12] *Surreal and Surcomplex Numbers in Symbolic Computer Algebra*, research report, same snapshot. [Report README](#); [LaTeX source](#). Used for the distinction between denotation, effective coefficient access, decision procedures and certified approximation. AI-assisted, unrefereed repository draft.